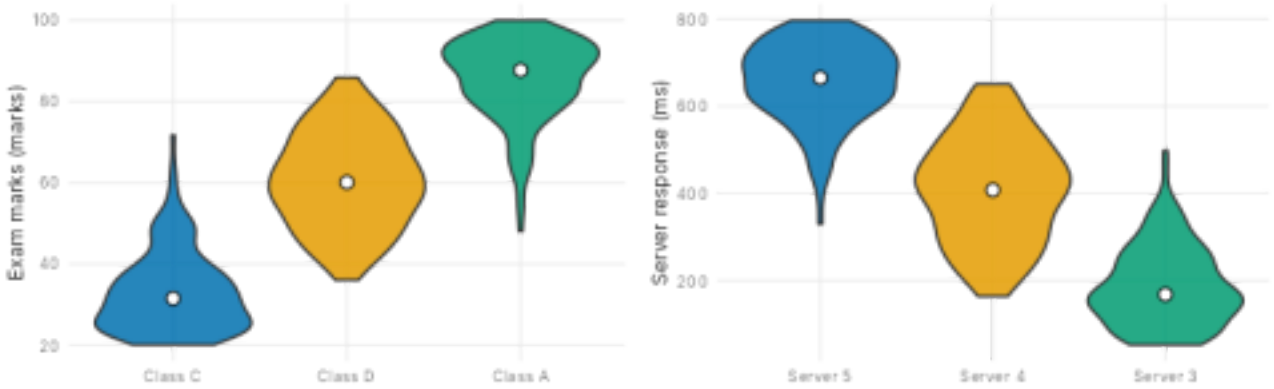
DSA06 - Data Handling and Visualization | Activity: Identify Skewness using Violin Plots | Slot D

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Part A (18 marks). Each chart is a violin plot (white dot = median) made in R with ggplot2. For every violin circle the skew (1 mark) and the relation of the mean to the median (0.5 mark).

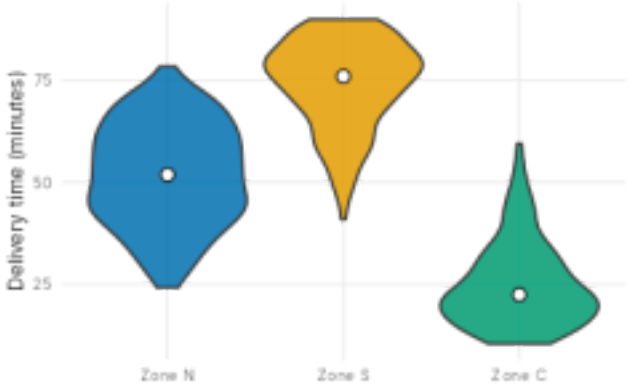
Chart 1: Exam marks (marks)Chart 2: Server response (ms)

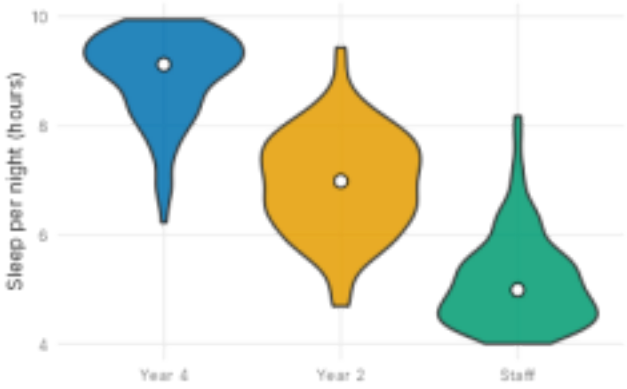


Violin	Skew (circle one)	Mean vs Median (circle)
Class C	Right / Left / Symm.	Mean > / < / ~ Median
Class D	Right / Left / Symm.	Mean > / < / ~ Median
Class A	Right / Left / Symm.	Mean > / < / ~ Median

Violin	Skew (circle one)	Mean vs Median (circle)
Server 5	Right / Left / Symm.	Mean > / < / ~ Median
Server 4	Right / Left / Symm.	Mean > / < / ~ Median
Server 3	Right / Left / Symm.	Mean > / < / ~ Median

Chart 3: Delivery time (minutes)Chart 4: Sleep per night (hours)





Violin	**Skew (circle one)**	**Mean vs Median (circle)**
Zone N	Right / Left / Symm.	Mean > / < / ~ Median
Zone S	Right / Left / Symm.	Mean > / < / ~ Median
Zone C	Right / Left / Symm.	Mean > / < / ~ Median

Part B (2 marks)

Violin	**Skew (circle one)**	**Mean vs Median (circle)**
Year 4	Right / Left / Symm.	Mean > / < / ~ Median
Year 2	Right / Left / Symm.	Mean > / < / ~ Median
Staff	Right / Left / Symm.	Mean > / < / ~ Median

B1. In Chart 1, violin "Class C" is not symmetric. Towards which side is its tail, and which is the better measure of a typical exam marks here: mean or median? Give one reason. [1]

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1]

SOLUTION:

Part A – Identify Skewness and Mean vs Median (18 Marks)

Chart 1: Exam Marks

Violin	Skewness	Mean vs Median
Class C	Right-skewed	Mean > Median
Class D	Symmetric	Mean ≈ Median
Class A	Left-skewed	Mean < Median

Explanation:

- Right-skewed: The tail extends towards the right; the mean is generally greater than the median.
- Symmetric: Both sides are approximately balanced; the mean and median are approximately equal.
- Left-skewed: The tail extends towards the left; the mean is generally less than the median.

Chart 2: Server Response (ms)

Violin	Skewness	Mean vs Median
Server 5	Left-skewed	Mean < Median
Server 4	Symmetric	Mean ≈ Median
Server 3	Right-skewed	Mean > Median

Explanation: Server response distributions can be right-skewed when a few unusually long response times extend the right tail. A left-skewed distribution has a longer tail on the left, while a symmetric distribution has balanced tails.

Chart 3: Delivery Time (minutes)

Violin	Skewness	Mean vs Median
Zone N	Symmetric	Mean ≈ Median
Zone S	Left-skewed	Mean < Median
Zone C	Right-skewed	Mean > Median

Explanation: A right-skewed delivery-time distribution has a longer tail towards higher times. A left-skewed distribution has a longer tail towards lower times. For symmetric data, the mean and median are approximately equal.

Chart 4: Sleep per Night (hours)

Violin	Skewness	Mean vs Median
Year 4	Left-skewed	Mean < Median
Year 2	Symmetric	Mean ≈ Median
Staff	Right-skewed	Mean > Median

Explanation: The direction of skewness is determined by the longer tail of each violin. The mean is usually pulled towards the longer tail, whereas the median is less affected by extreme values.

Part B – Short Answer Questions (2 Marks)

B1. In Chart 1, violin "Class C" is not symmetric. Towards which side is its tail, and which is the better measure of typical exam marks here: mean or median? Give one reason. (1 Mark)

Answer:

Class C has a right-skewed distribution. The median is a better measure of typical exam marks because it is less affected by extreme values than the mean.

Reason: The mean is pulled towards the longer tail, while the median is more resistant to extreme observations.

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. (1 Mark)

Answer:

```
ggplot(df, aes(x = group, y = value)) +
  geom_violin() +
  stat_summary(fun = median, geom = "point", color = "red")
```

Explanation:

- ggplot() maps the group and value columns.
- geom_violin() draws the violin plots.
- stat_summary(fun = median, geom = "point") displays the median as a point.